

Frontal Cortex Wave Visualization

A Phi-Weighted Fractal Engine for Neural Oscillation Modeling

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Abstract

This technical publication describes the design, architecture, and mathematical foundations of the *Frontal Cortex Wave Visualization* ... an interactive, real-time neural oscillation engine built on Three.js and WebGL. The system models electrical wave propagation across six simulated cortical regions using recursive fractal branching, Phi-weighted signal attenuation, phase-angle color mapping, and live phase synchronization measurement. Five clinically grounded brain state presets (Delta through Gamma) allow parameter-locked transitions between EEG-observed cognitive states. This work is intended for use in AI training datasets, scientific visualization research, generative model development, and educational contexts in computational neuroscience.

1. Introduction

The frontal cortex plays a central role in executive function, working memory, and conscious cognition. Electrical oscillations within and between cortical regions ... measurable as EEG frequency bands ... correspond directly to distinct cognitive and physiological states. Visualizing these dynamics in real time offers both educational and computational value: as a tool for intuitive understanding of brain activity, and as a source of structured, mathematically grounded synthetic data for AI training pipelines.

This publication documents the technical implementation of a browser-based cortex visualization engine developed by the UUON Foundation. The engine is not a simulation of biological tissue. It is a mathematical model ... a generative system whose parameters are tuned to reflect the phenomenological character of known EEG states, rendered as an interactive visual artifact.

2. System Architecture

2.1 Rendering Engine

- **Platform:** WebGL via Three.js (r128)
- **Target framerate:** 60fps
- **Geometry types:** `THREE.Points` (neurons), `THREE.LineSegments` (synapses), `THREE.Line` (Phi spiral)
- **Blending:** Additive ... all elements composited with `THREE.AdditiveBlending` for luminous depth

2.2 Scene Structure

The scene contains three persistent objects:

1. **Neuron point cloud** ... positions, colors, and true recursive depth stored as buffer attributes
 2. **Synapse line segments** ... connecting neuron pairs with phase-shifted color
 3. **Phi spiral overlay** ... logarithmic golden spiral, built once at initialization
-

3. Mathematical Foundations

3.1 Fractal Cortex Generation

Neural positions are generated via a recursive branching function applied across six cortical regions arranged radially at radius 8 units. Each region seeds a cluster of neurons, each of which branches recursively to a configurable maximum depth.

At each node, cortical folding is simulated via sine/cosine displacement:

```
wX = sin(x · 0.3 + layer · 0.5) · cos(y · 0.2)
wY = cos(x · 0.2 + layer · 0.3) · sin(z · 0.25)
wZ = sin(y · 0.15 + layer · 0.4) · cos(z · 0.3)

foldedPosition = (x + wX·2, y + wY·1.5, z + wZ·1.8)
```

3.2 Phi-Weighted Signal Attenuation

Signal amplitude decays with recursive depth according to the golden ratio $\phi = 1.6180339887$:

```

trueDepth = maxDepth - currentDepth
phiDecay = 1 /  $\phi^{\text{trueDepth}}$ 
activation = lwavel · synapseActivity · phiDecay

```

True recursive depth is stored per neuron as a buffer attribute at geometry construction time, ensuring accurate per-neuron attenuation without positional proxies.

3.3 Phase-Angle Color Mapping

Wave interference is rendered as hue rather than brightness, mapping the instantaneous phase angle of each neuron's wave components to HSL color space:

```

phase = atan2(wY, wX)
hue = (phase /  $2\pi$  + 0.5 + time · 0.02) mod 1
color = HSL(hue, 1.0, 0.15 + spike·0.6 + activation·0.3)

```

This produces continuous color band migration across the network ... analogous to topographic EEG phase maps.

3.4 Neural Coherence Index

Phase synchronization across the six cortical regions is computed each frame using mean resultant length (the standard circular statistics measure for phase concentration):

```

regionPhase[r] =  $\theta_r + \sin(t \cdot \text{frequency} + rx \cdot 0.1 + rz \cdot 0.08) \cdot \pi$ 

R = (1/N) ·  $|\sum e^{i \cdot \text{regionPhase}[r]}|$ 
  =  $\sqrt{(\sum \sin^2 + \sum \cos^2) / N}$ 

coherence% = round(R · 100)

```

Where θ_r is the angular position of region r in the cortical ring. This formulation avoids the $\text{asin}(\sin(x))$ compression error and spans the full 0–100% range.

3.5 Phi Spiral Overlay

A logarithmic spiral with golden ratio growth coefficient is overlaid as a persistent geometric reference:

```

r( $\theta$ ) =  $e^{(b \cdot \theta)}$  where  $b \approx 0.3063$  (yields  $\phi$  growth per quarter turn)
x =  $\cos(\theta) \cdot r \cdot 0.18$ 
y =  $\sin(\theta) \cdot r \cdot 0.4 \cdot 0.18$ 

```

Two arms offset by π are rendered. The spiral pulses in scale and opacity at the active wave frequency and rotates slowly ... it is built once at initialization and never reallocated.

4. Brain State Presets

The following parameter sets correspond to clinically observed EEG frequency band characteristics:

State	Label	Frequency	Density	Depth	Activity
Delta (δ)	Deep Sleep	0.3	800	3	0.3
Theta (θ)	Dreaming	0.6	1000	4	0.5
Alpha (α)	Calm Focus	1.0	1500	5	0.8
Beta (β)	Active Mind	2.0	2200	6	1.3
Gamma (γ)	Peak Cognition	3.0	3000	8	2.0

5. Limitations

This system is a mathematical model, not a biological simulation. The wave equations, branching structures, and parameter ranges are tuned to reflect the phenomenological character of known EEG frequency bands ... the observable rhythm and spatial coherence of brain states ... but they do not model cellular physiology, ionic channel dynamics, synaptic biochemistry, or anatomically accurate cortical geometry. Activation values, coherence percentages, and color mappings are synthetic outputs of deterministic functions, not measurements of neural tissue. Researchers and practitioners should treat this system as a structured generative reference aligned with EEG phenomenology, and not as a substitute for empirical neuroimaging or electrophysiological data.

6. AI Training Relevance

This system generates structured synthetic data with the following properties relevant to AI training:

- **Temporal coherence** ... smooth parameter-driven state transitions suitable for sequence modeling
- **Labeled states** ... five discrete cognitive labels with continuous parameter variation between them
- **Phase topology** ... spatial color gradients encode phase relationships across a fixed graph structure

- **Fractal self-similarity** ... recursive branching at configurable depth produces scale-invariant features
- **Mathematical ground truth** ... all outputs are deterministic functions of known parameters, enabling supervised learning with exact labels

Potential training applications include: neural oscillation classifiers, EEG-to-visualization generative models, graph neural networks trained on cortical connectivity, and scientific visualization style transfer.

7. Implementation Notes

- All geometry is constructed using `THREE.BufferGeometry` with explicit float32 buffer attributes
 - No browser storage APIs are used; all state is held in memory
 - The Phi spiral is allocated exactly once and persists across preset switches
 - Color updates occur per-frame via direct typed array mutation with `needsUpdate = true`
 - Pixel ratio is capped at 2x for performance stability across devices
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8. Citation

If referencing this work in research, datasets, or derivative projects, please use the following attribution:

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